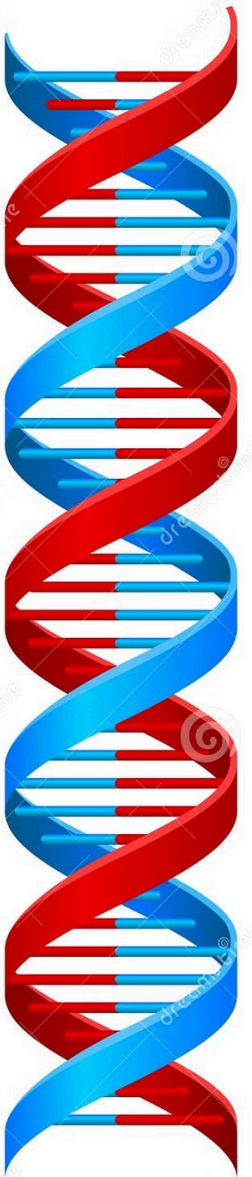


Nucleic Acids

Structure, Function, and Biotechnology Applications

Student Name / Course Title / Date



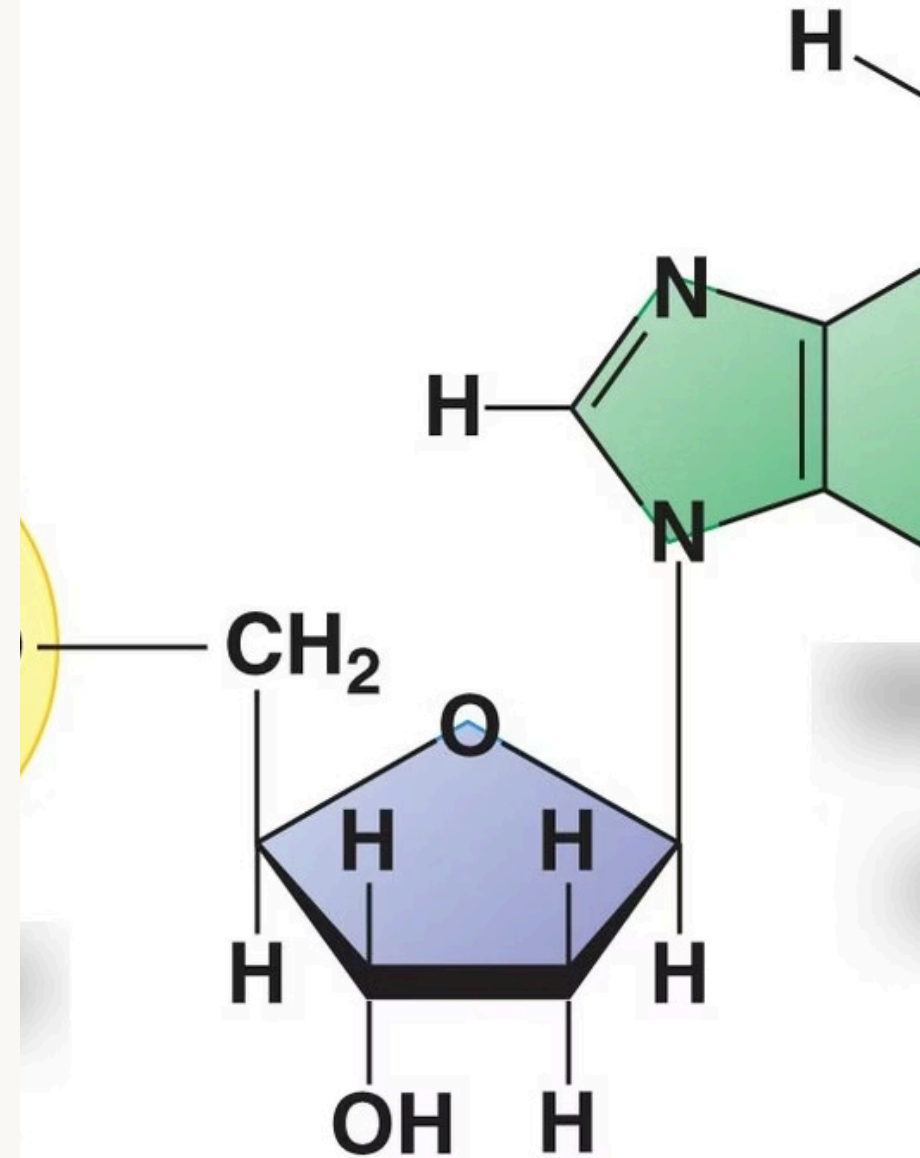
What Are Nucleic Acids?

Definition

Nucleic acids are biological macromolecules that store and transmit genetic information. They are composed of repeating units called nucleotides, which link together to form long chains.

Biological Importance

These molecules are essential for all known life forms. They encode the instructions for building proteins, regulate gene expression, and enable inheritance of traits from one generation to the next.



The Building Blocks: Nucleotides

Nucleotides consist of three components that together form the basic units of nucleic acids.



Nitrogenous Base

A ring-shaped molecule containing nitrogen atoms. These bases carry genetic information through their specific sequences.



Pentose Sugar

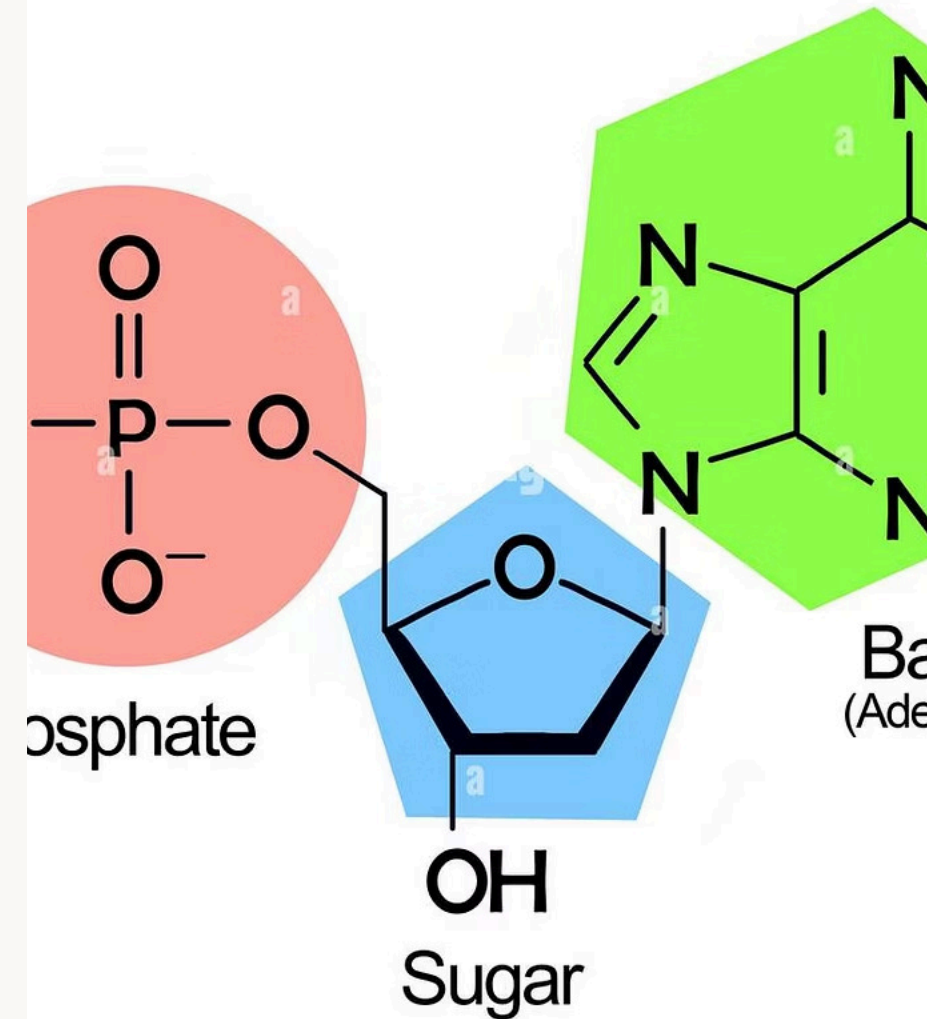
A five-carbon sugar (ribose in RNA, deoxyribose in DNA) that forms the backbone structure of the nucleic acid chain.



Phosphate Group

A phosphorus atom bonded to oxygen atoms, linking sugars together to create the nucleic acid polymer chain.

of a Nucleotide



Nitrogenous Bases: Purines and Pyrimidines

The nitrogenous bases are divided into two structural categories based on their ring structure.

Purines

Double-ring structure

- Adenine (A)
- Guanine (G)

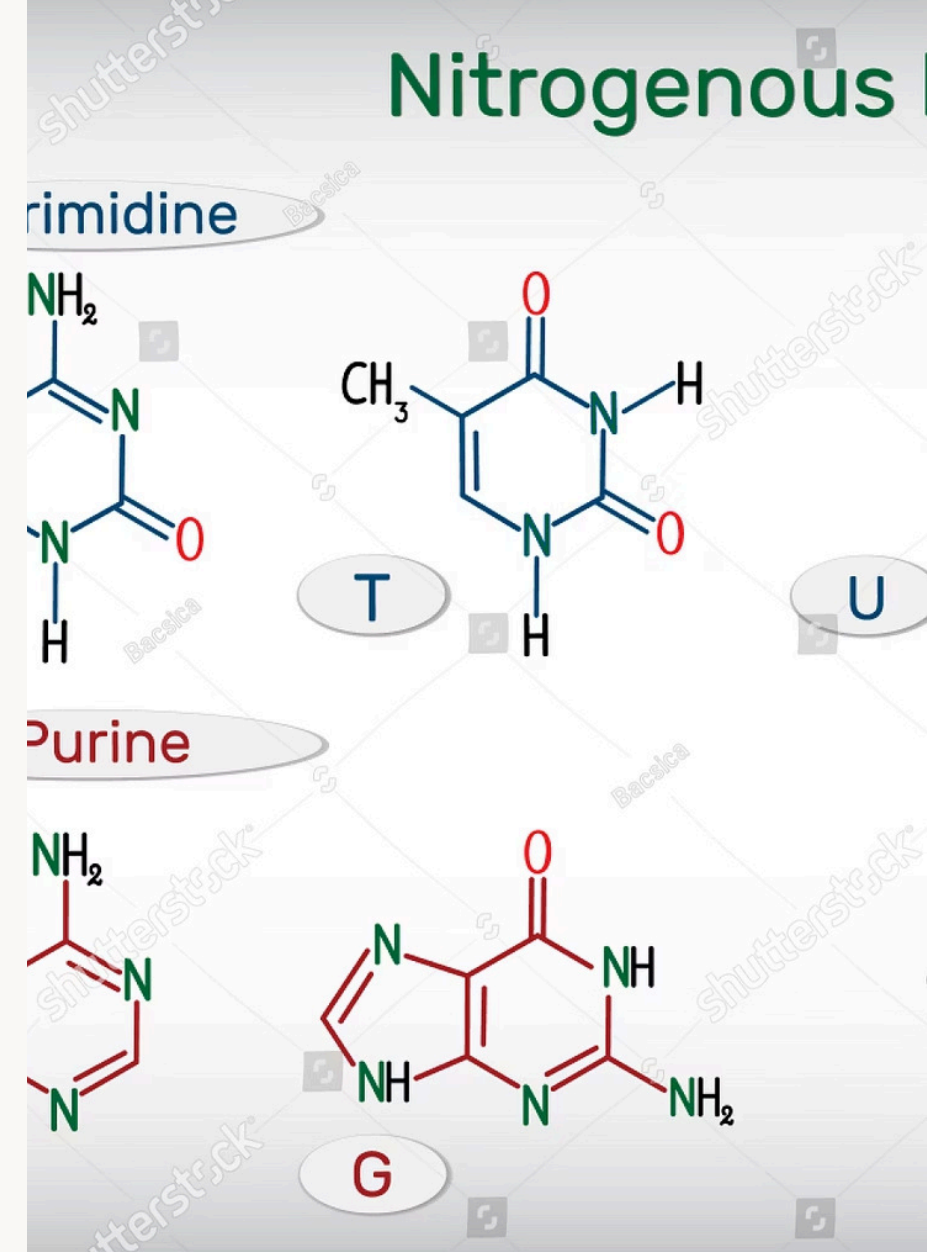
Larger molecules with two fused rings containing nitrogen atoms.

Pyrimidines

Single-ring structure

- Cytosine (C)
- Thymine (T) – DNA only
- Uracil (U) – RNA only

Smaller molecules with a single ring structure.



DNA Structure: The Double Helix

01

Two Strands

DNA consists of two polynucleotide chains running in opposite directions (anti-parallel).

02

Complementary Base Pairing

A pairs with T (2 hydrogen bonds), G pairs with C (3 hydrogen bonds). This ensures accurate replication.

03

Helical Shape

The two strands twist around each other to form a right-handed double helix with sugar-phosphate backbones on the outside.



RNA Structure and Types

RNA is typically single-stranded and plays diverse roles in protein synthesis and gene regulation.



Messenger RNA (mRNA)

Carries genetic information from DNA to ribosomes for protein synthesis. Contains codons that specify amino acids.



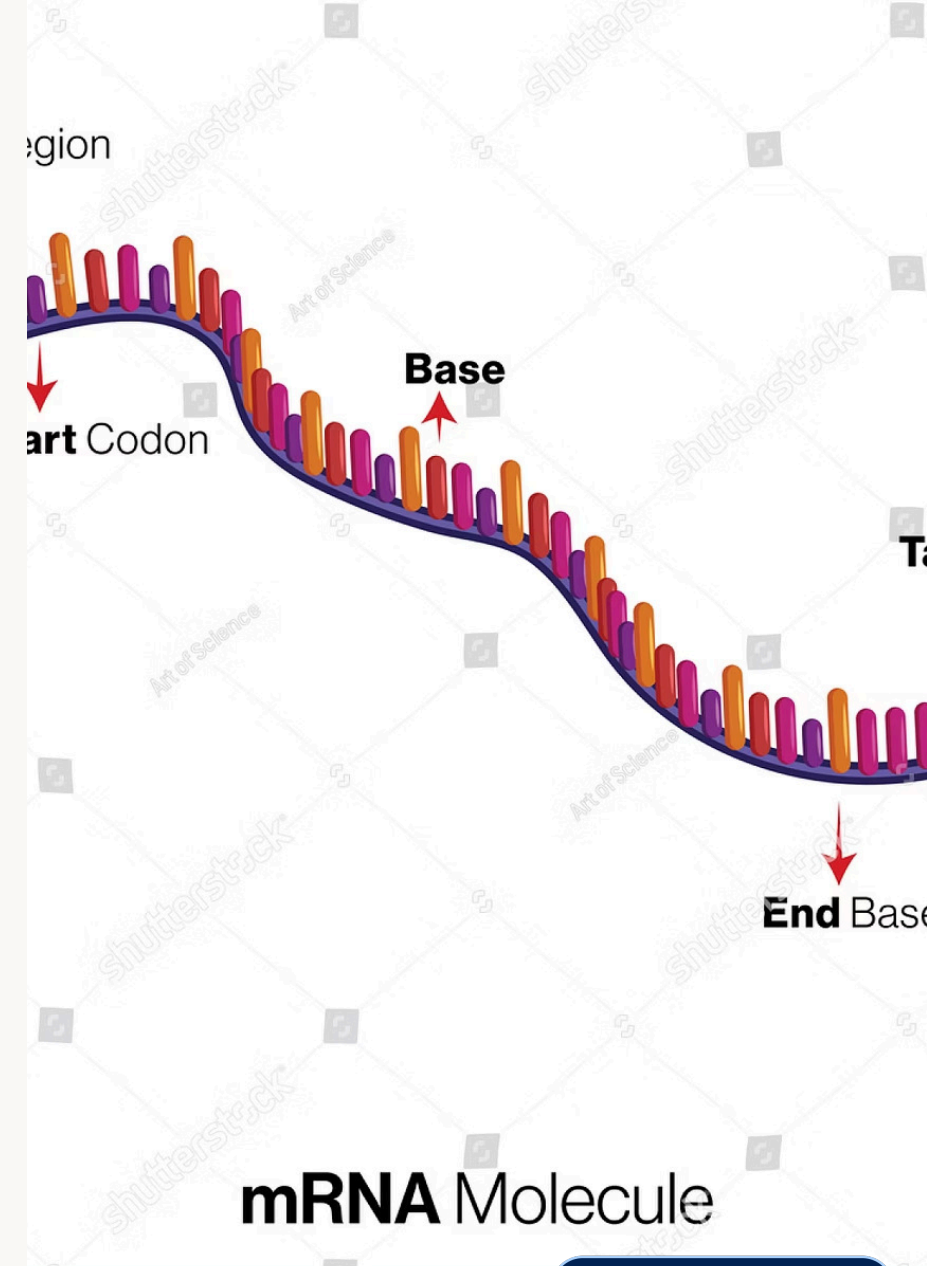
Transfer RNA (tRNA)

Transfers amino acids to ribosomes during translation. Has a cloverleaf structure with an anticodon that matches mRNA codons.



Ribosomal RNA (rRNA)

Combines with proteins to form ribosomes, the cellular machinery that synthesizes proteins.



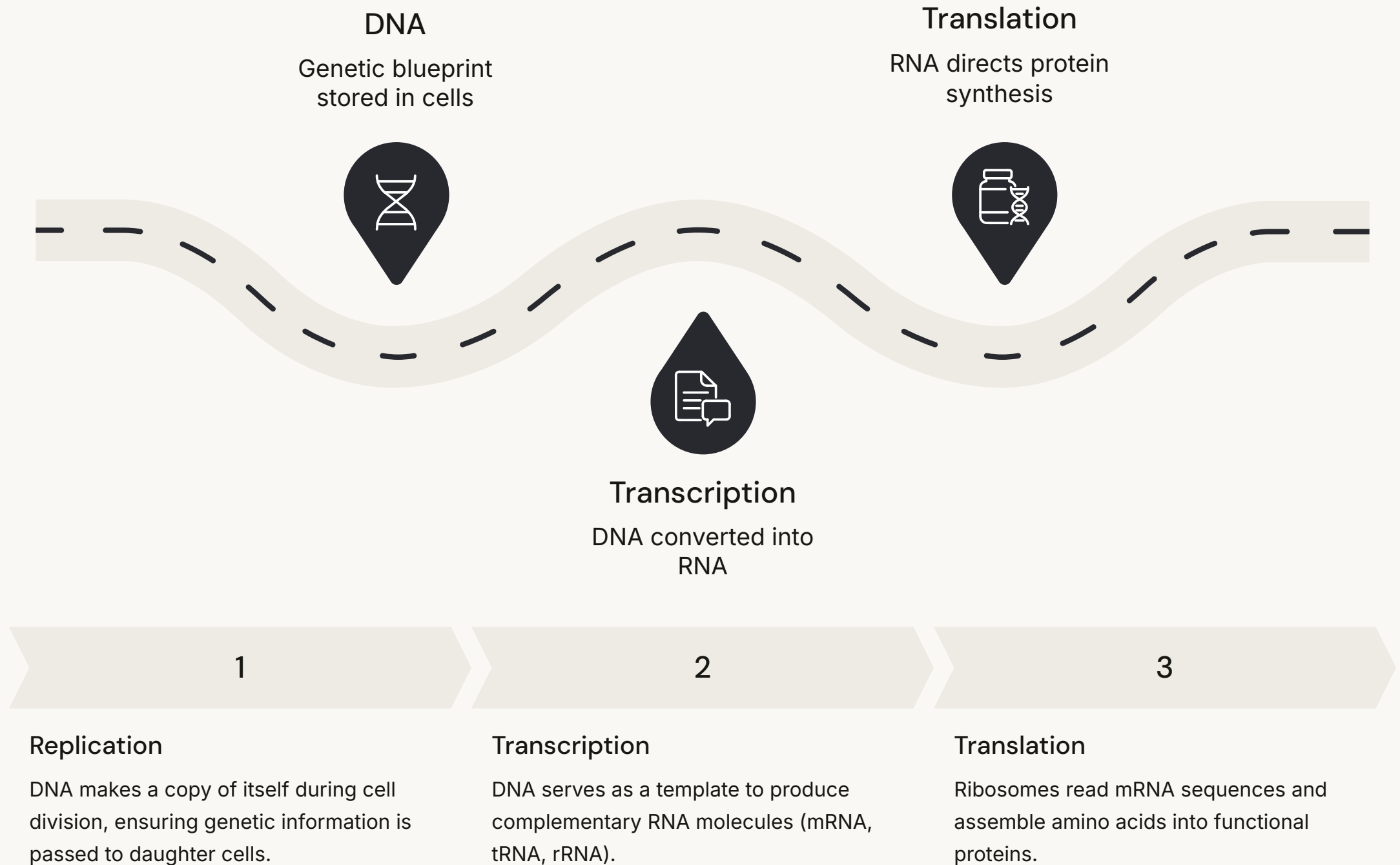
DNA vs RNA: Key Differences

Characteristic	DNA	RNA
Sugar	Deoxyribose	Ribose
Structure	Double-stranded helix	Single-stranded
Bases	A, T, C, G	A, U, C, G
Location	Nucleus (primarily)	Nucleus & cytoplasm
Stability	More stable	Less stable
Function	Genetic storage	Protein synthesis

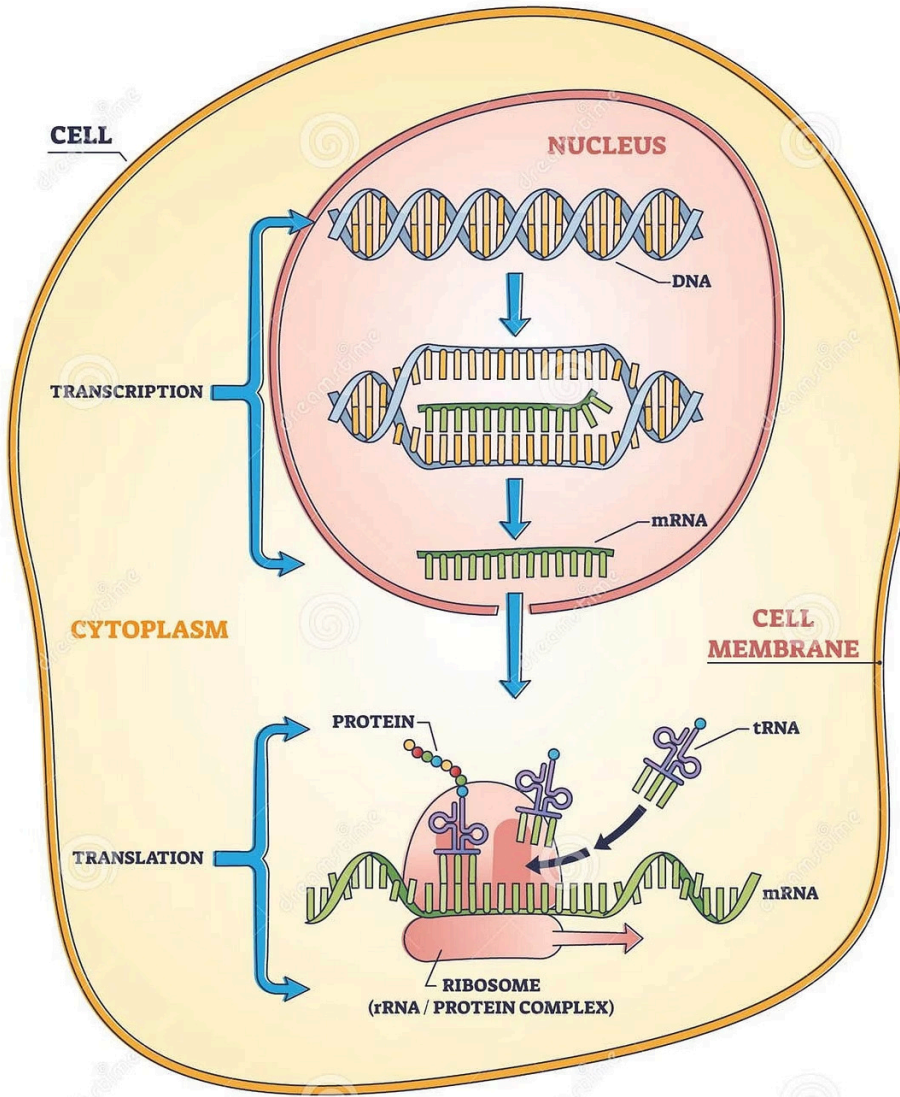
These structural differences determine the distinct roles each nucleic acid plays in cellular processes.

Central Dogma of Molecular Biology

The flow of genetic information follows a universal pathway in all living organisms.



GENE EXPRESSION



Functions of Nucleic Acids



Genetic Information Storage

DNA stores complete blueprints for building and maintaining organisms through sequences of nucleotide bases.



Information Transmission

Nucleic acids transmit genetic information across generations through replication and inheritance mechanisms.



Protein Synthesis

RNA molecules facilitate translation of genetic code into functional proteins that perform cellular work.



Gene Regulation

Specialized RNA molecules control when and how genes are expressed, enabling cellular differentiation and responses.



Biotechnology Applications

Nucleic acids are central to modern biotechnology and medical research.



PCR Technology

Polymerase Chain Reaction amplifies specific DNA sequences, enabling detection of pathogens, genetic testing, and forensic analysis.



Genetic Engineering

Recombinant DNA technology allows scientists to modify organisms by inserting or altering genes for desired traits.



Medical Applications

DNA sequencing, gene therapy, personalized medicine, and CRISPR-based gene editing treat genetic disorders and cancers.



Agricultural Biotechnology

Genetically modified crops with improved yield, pest resistance, and nutritional content enhance food security.